

HIDDEN GEMS

A YouTube Discovery Platform Initiative

FTRD

Hidden Gems — Niche Creator Discovery Platform

Functional & Technical Requirements Document

Project	Hidden Gems Discovery Layer (YouTube)
Document Version	1.0
Status	Draft for Engineering Review
Date	July 14, 2026
Prepared By	Business Analysis / Engineering
Classification	Internal — Product & Engineering Confidential

Document Control

Revision History

Version	Date	Author	Description of Change
0.1	2026-06-25	Business Analyst	Initial functional decomposition from BRD
0.6	2026-07-05	Solutions Architect	Added technical architecture & NFRs
1.0	2026-07-14	Business Analyst	Baselined for engineering sign-off

Approval Sign-Off

Role	Name	Signature	Date
Engineering Lead, Recommendations Platform			
Engineering Lead, Creator Studio			
Solutions Architect			
QA Lead			
Product Manager			

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1. Introduction

1.1 Purpose

This Functional & Technical Requirements Document (FTRD) translates the business requirements defined in the Hidden Gems Business Requirements Document (BRD v1.0) into detailed functional requirements (FR), non-functional requirements (NFR), and a high-level technical solution design. It is the authoritative specification used by engineering, QA, and architecture to design, build, and test the Hidden Gems discovery layer.

1.2 Audience

- Engineering teams (Creator Studio, Recommendations Platform, Search, Analytics)
- Solutions and enterprise architects
- Quality assurance and test engineering
- Product management and business analysis

1.3 Requirement Numbering Convention

Functional requirements are numbered FR-<Module>-<Seq> and grouped by functional module. Non-functional requirements are numbered NFR-<Seq>. Every requirement traces to one or more Business Requirements (BR-xx) from the BRD; the full mapping is provided in the Requirements Traceability Matrix (RTM) in the Supporting Artifacts document.

2. Functional Modules Overview

Module	Name	Description
M1	Creator Labeling	Workflow enabling creators to attach, edit, and submit niche descriptor labels during upload/publish.
M2	Label Verification (AI)	Automated verification service validating labels against taxonomy and content, with human escalation.
M3	Discovery Surface (Viewer)	Home-tab surface, niche chips, and card rendering for Hidden Gems content.
M4	Search by Intent	Viewer search experience allowing direct query by niche/intent label.
M5	Recommendation & Ranking Engine	Behavioral AI scoring and ranking of eligible videos per viewer.
M6	Watch Page & Creator Profile Integration	Surfacing of Hidden Gems content and badges on watch page and profile.
M7	Attribution & Analytics	Tracking, attribution, and reporting of incremental visits and revenue.
M8	Trust & Safety / Moderation	Review queue and controls for flagged or abusive labels.

3. Functional Requirements

3.1 Module M1 — Creator Labeling

FR ID	Requirement	Priority	Traced BR
FR-M1-01	The system shall allow a creator to add one or more niche descriptor labels to a video during the upload flow prior to publishing.	Must	BR-01
FR-M1-02	The system shall suggest candidate labels to the creator based on video title, description, and content analysis.	Should	BR-01
FR-M1-03	The system shall allow a creator to enter a free-text custom label in addition to suggested labels.	Must	BR-01
FR-M1-04	The system shall display estimated reach uplift messaging (e.g., “adding labels improves reach by ~3x”) to encourage labeling.	Should	BR-01
FR-M1-05	The system shall allow a creator to edit or remove labels on an already-published video from Creator Studio.	Must	BR-01
FR-M1-06	The system shall persist label metadata against the video record and version history.	Must	BR-01

3.2 Module M2 — Label Verification (AI)

FR ID	Requirement	Priority	Traced BR
FR-M2-01	The system shall automatically evaluate every submitted label against an approved niche taxonomy and the video's content signals prior to publish.	Must	BR-02
FR-M2-02	The system shall reject or flag labels identified as spam, misleading, or abusive, and shall notify the creator with a reason.	Must	BR-02
FR-M2-03	The system shall route ambiguous or borderline label decisions to a human moderation queue (Module M8).	Must	BR-02
FR-M2-04	The system shall maintain and version the approved niche label taxonomy, allowing controlled expansion as new labels are approved.	Should	BR-02
FR-M2-05	The system shall log verification decisions and confidence scores for audit and model-retraining purposes.	Must	BR-02

3.3 Module M3 — Discovery Surface (Viewer)

FR ID	Requirement	Priority	Traced BR
FR-M3-01	The system shall render a “Hidden Gems” entry point on the YouTube home tab for eligible viewers.	Must	BR-03

FR ID	Requirement	Priority	Traced BR
FR-M3-02	The system shall display niche preference chips (e.g., Cinematography, Chess, Slow Living) allowing the viewer to filter the surface.	Must	BR-03
FR-M3-03	The system shall render video cards labeled “Hidden Gem” with creator name, subscriber tier, and view count.	Must	BR-03
FR-M3-04	The system shall refresh the Hidden Gems surface with new eligible content on a defined cadence to ensure freshness.	Should	BR-03
FR-M3-05	The system shall allow a viewer to dismiss or provide negative feedback on a surfaced Hidden Gem to improve future ranking.	Could	BR-03

3.4 Module M4 — Search by Intent

FR ID	Requirement	Priority	Traced BR
FR-M4-01	The system shall allow a viewer to search using a niche/intent label and return matching Hidden Gems content ranked above generic search results where relevant.	Must	BR-04
FR-M4-02	The system shall support autocomplete/suggestion of known taxonomy labels as the viewer types a search query.	Should	BR-04
FR-M4-03	The system shall visually distinguish Hidden Gems results from standard search results with consistent badging.	Must	BR-04

3.5 Module M5 — Recommendation & Ranking Engine

FR ID	Requirement	Priority	Traced BR
FR-M5-01	The system shall score candidate videos using label affinity, viewer watch history, and content freshness signals.	Must	BR-03, BR-04
FR-M5-02	The system shall exclude videos from creators exceeding the defined subscriber ceiling (500,000) from Hidden Gems eligibility.	Must	BR-03
FR-M5-03	The system shall support A/B experimentation on ranking weights without requiring a client release.	Must	BR-08
FR-M5-04	The system shall re-rank the surface in near real time as new viewer engagement signals are received.	Should	BR-03
FR-M5-05	The system shall meet a minimum recommendation relevance target of 85% as measured by the defined relevance evaluation methodology.	Must	BR-10

3.6 Module M6 — Watch Page & Creator Profile Integration

FR ID	Requirement	Priority	Traced BR
FR-M6-01	The system shall display a “Featured in Hidden Gems” badge on the watch page for eligible videos.	Should	BR-05
FR-M6-02	The system shall surface related Hidden Gems (same label, same creator) in the watch page's related-content module.	Should	BR-05
FR-M6-03	The system shall provide a direct link from a Hidden Gems card or watch page to the creator's profile.	Must	BR-06
FR-M6-04	The system shall display the creator's active niche labels and related Hidden Gems on their profile page.	Should	BR-06

3.7 Module M7 — Attribution & Analytics

FR ID	Requirement	Priority	Traced BR
FR-M7-01	The system shall attribute a profile visit, subscribe, or watch session to Hidden Gems when the originating impression came from the surface.	Must	BR-07
FR-M7-02	The system shall attribute ad, membership, and Super Chat revenue events to Hidden Gems using the same attribution window and rules.	Must	BR-07
FR-M7-03	The system shall expose a creator-facing analytics dashboard showing Hidden Gems views, CTR, and attributed revenue.	Should	BR-09
FR-M7-04	The system shall expose an internal reporting layer rolling up the North Star Metric and L1/L2 supporting metrics daily.	Must	BR-07

3.8 Module M8 — Trust & Safety / Moderation

FR ID	Requirement	Priority	Traced BR
FR-M8-01	The system shall provide a moderation queue for labels flagged by the AI verification service.	Must	BR-02
FR-M8-02	The system shall allow a moderator to approve, reject, or edit a flagged label, with the decision logged.	Must	BR-02
FR-M8-03	The system shall allow moderators to suspend a creator's labeling privileges for repeated policy violations.	Should	BR-02

4. Non-Functional Requirements

NFR ID	Category	Requirement	Target
NFR-01	Availability	The Hidden Gems surface and APIs shall be available during peak traffic periods.	≥ 99% uptime
NFR-02	Performance	Discovery surface and search-by-intent responses shall render within the defined latency budget.	p95 < 400 ms
NFR-03	AI Quality	The label verification model shall correctly classify valid vs. invalid labels.	≥ 90% precision
NFR-04	Recommendation Quality	Ranked recommendations shall be judged relevant by the evaluation methodology.	≥ 85% relevance
NFR-05	Scalability	The platform shall support the target viewer segment and mid-tier creator population without service degradation.	See §5.3 capacity model
NFR-06	Security & Privacy	All viewer preference and watch-history signals used for ranking shall comply with existing data-privacy and consent policies.	100% policy compliance
NFR-07	Auditability	All label verification and moderation decisions shall be logged with timestamp, actor, and rationale for a minimum retention period.	≥ 13 months retention
NFR-08	Experimentation	Ranking and surface placement changes shall be deployable via A/B experiment configuration without a client app release.	N/A
NFR-09	Accessibility	All new UI surfaces shall meet WCAG 2.1 AA accessibility standards.	WCAG 2.1 AA
NFR-10	Internationalization	Labeling taxonomy and discovery surface shall support localization for priority launch markets.	Per market rollout plan

5. Technical Solution Overview

5.1 Logical Architecture

Hidden Gems is delivered as a layer on top of existing YouTube platform services rather than a parallel system. The solution introduces four new/extended services:

Component	Responsibility
Labeling Service (extends Creator Studio)	Captures, validates format, and persists creator-submitted labels; surfaces suggested labels.
Label Verification Service (new, AI/ML)	Classifies labels against taxonomy and content signals; routes exceptions to moderation.
Discovery Ranking Service (extends Recommendations Platform)	Scores and ranks eligible Hidden Gems candidates per viewer using affinity, history, and freshness features.
Attribution Service (new, extends Analytics Platform)	Tracks impressions/clicks from Hidden Gems surfaces and attributes downstream visits and revenue events.

Client surfaces (Home tab, Search, Watch Page, Creator Profile, Creator Studio) consume these services via existing platform API gateways; no new client applications are introduced.

5.2 Data Requirements

Entity	Key Attributes	Notes
Label	label_id, taxonomy_ref, display_text, status, created_by	Status: pending approved rejected
Video-Label Association	video_id, label_id, verification_score, verified_at	Many-to-many; supports multiple labels per video
Viewer Preference Profile	viewer_id, affinity_vector, opted_labels, last_updated	Derived from watch history and explicit chip selection
Discovery Impression	impression_id, viewer_id, video_id, surface, rank_position, ts	Powers ranking evaluation and attribution
Attribution Event	event_id, impression_id, event_type, value, attributed_at	event_type: visit subscribe ad_revenue membership superchat

5.3 Capacity & Scale Considerations

- Design for the full addressable mid-tier creator population and target high-intent viewer segment sized in the Business Impact model (see BRD §2 and Supporting Artifacts § Business Impact Assumptions).
- Ranking service must support near-real-time re-scoring as engagement signals arrive without breaching the p95 < 400ms latency guardrail.
- Verification service throughput must scale with creator beta onboarding (initial cohort of 5,000 creators in Phase 1) without introducing publish-flow delays.

5.4 Integration Requirements

- Creator Studio upload/publish flow ↔ Labeling Service (synchronous, blocking on verification result before public labels go live).
- Recommendations Platform ↔ Discovery Ranking Service (feature store integration for affinity and freshness signals).
- Analytics Platform ↔ Attribution Service (event stream integration for impressions, clicks, and revenue events).
- Trust & Safety moderation tooling ↔ Label Verification Service (queue and case-management integration).

6. Traceability

A complete requirement-to-requirement traceability matrix (Business Objective → Business Requirement → Functional Requirement → User Story → Test Condition) is maintained in the Supporting Artifacts document, Section “Requirements Traceability Matrix (RTM).”